

Homework 1

PART 1

Task: Operating System Concepts Chapter 1 Exercises: 1.14, 1.17, 1.19, 1.22

Q 1

1.14 What is the purpose of interrupts? How does an interrupt differ from a trap? Can traps be generated intentionally by a user program? If so, for what purpose?

Answer: Interrupts are used to stop what the CPU is doing and immediately transfers execution to a fixed location. With this action we can realize communication between controller and device driver. An interrupt occur, there is a fixed location to go and handle this situation.

While a interrupt realizes communication between controller and device driver, a trap is mainly used to handle the communication between CPU and memory, containing the fault caused by calculating. I list several differences between them.

1. Traps are usually caused by current instructions being executed by the processor, while interrupts are caused by interrupt sources unrelated to the current instructions.
2. The service provided by the trap handler is used by the current process, while the service provided by the interrupt handler is not for the current process.
3. After CPU executes an instruction, it responds to interrupts before the next instruction starts, and it can also respond to traps during the execution of an instruction.

Yes, traps can be generated by a user program, it can be used for debugging.

Q 2

1.17 Some computer systems do not provide a privileged mode of operation in hardware. Is it possible to construct a secure operating system for these computer systems? Give arguments both that it is and that it is not possible.

Answer:

Possible: Comparing with the system providing a privileged mode, it may be safer as some instruction can't execute in user mode. And computer systems not providing a privileged mode can avoid the unnecessary access, so the user can not access and modify the system file, it will helpful to avoid some attacks.

Not possible: As the privileged mode is closed in hardware, computer users containing administrator can not do the privileged operation, it will lead to a problem that professional user can't utilize their ability to solve some defects and loopholes. It will also bring unsecure situation.

Q 3

1.19 Rank the following storage systems from slowest to fastest:

- a. Hard-disk drives
- b. Registers
- c. Optical disk
- d. Main memory
- e. Nonvolatile memory
- f. Magnetic tapes
- g. Cache

Answer: f < c < a < e < d < g < b

Q 4

1.22 Describe a mechanism for enforcing memory protection in order to prevent a program from modifying the memory associated with other programs.

Answer: We can add an item called program ID in the page table, every program has its own it, when the access request from the corresponding program can be admitted. It can avoid different program access the same part of memory.

PART 2

Task: Detail my steps about how to get arch/arm64/kernel/sys.i

Answer: In order to get the target file, there are several steps to do. (*The specific command is given under every step.)

1. Install arm64 cross compile tool.

```
sudo apt-get install gcc-aarch64-linux-gnu -y
```

2. Generate the config file.

```
make ARCH=ARM64 CROSS_COMPILE=aarch64-linux-gnu defconfig
```

```
percy@percy-P65-67RSRP:~/kernel/linux-5.3.1$ make ARCH=arm64 CROSS_COMPILE=aarch64-linux-gnu- defconfig
*** Default configuration is based on 'defconfig'
#
# configuration written to .config
#
```

3. Compile.

```
make ARCH=ARM64 CROSS_COMPILE=aarch64-linux-gnu defconfig
arch/arm64/kernel/sys.i
```

```

percy@percy-P65-67RSRP:~/kernel/linux-5.3.1$ make ARCH=arm64 CROSS_COMPILE=aarch64-linux-gnu- arch/arm64/kernel/sys.i
scripts/kconfig/conf --syncconfig Kconfig
arch/arm64/Makefile:56: CROSS_COMPILE_COMPAT not defined or empty, the compat vDSO will not be built
make[1]: Warning: File 'scripts/mod/.sumversion.o.cmd' has modification time 16135 s in the future
CC      scripts/mod/empty.o
MKELF    scripts/mod/elfconfig.h
HOSTCC   scripts/mod/modpost.o
CC      scripts/mod/devicetable-offsets.s
HOSTCC   scripts/mod/file2alias.o
HOSTLD   scripts/mod/modpost
make[1]: warning: Clock skew detected. Your build may be incomplete.
make[1]: Warning: File 'arch/arm64/kernel/.asm-offsets.s.cmd' has modification time 15884 s in the future
CC      kernel/bounds.s
UPD      include/generated/bounds.h
CC      arch/arm64/kernel/asm-offsets.s
UPD      include/generated/asm-offsets.h
CALL     scripts/checksyscalls.sh
CALL     scripts/atomic/check-atomics.sh
make[1]: warning: Clock skew detected. Your build may be incomplete.
make[1]: Warning: File 'include/generated/.vdso-offsets.h.cmd' has modification time 15884 s in the future
make[1]: warning: Clock skew detected. Your build may be incomplete.
make[1]: Warning: File 'include/config/arch/dma/addr/t/64bit.h' has modification time 15882 s in the future
CPP      arch/arm64/kernel/sys.i
make[1]: warning: Clock skew detected. Your build may be incomplete.

```

```

percy@percy-P65-67RSRP:~/kernel/linux-5.3.1$ cd arch/arm64/kernel
percy@percy-P65-67RSRP:~/kernel/linux-5.3.1/arch/arm64/kernel$ ls | grep "sys.i"
sys.i
percy@percy-P65-67RSRP:~/kernel/linux-5.3.1/arch/arm64/kernel$

```

PART 3

Task: Find system call table for ARM32, RISC-V(32 bit), RISC-V(64 bit), X86(32 bit), X86_64 (50 points)

List source code file, the whole system call table with macro expanded.

ARM32

source code file: entry-common.S → entry-common.s

```

percy@percy-P65-67R5RP:~/kernel/linux-5.3.1/arch/arm/kernel$ cat entry-common.S
/* SPDX-License-Identifier: GPL-2.0-only */
/*
 * linux/arch/arm/kernel/entry-common.S
 *
 * Copyright (C) 2000 Russell King
 */

#include <asm/asm.h>
#include <asm/unistd.h>
#include <asm/ftrace.h>
#include <asm/unwind.h>
#include <asm/memory.h>
#ifdef CONFIG_AEABI
#include <asm/unistd-oabi.h>
#endif

        .equ      NR_syscalls, __NR_syscalls

#ifdef CONFIG_NEED_RET_TO_USER
#include <mach/entry-macro.S>
#else
        .macro     arch_ret_to_user, tmp1, tmp2
        .endm
#endif

#include "entry-header.S"

saved_psr        .req      r8
#ifdef CONFIG_TRACE_IRQFLAGS || CONFIG_CONTEXT_TRACKING
saved_pc          .req      r9
#define TRACE(x...) x
#else
saved_pc          .req      lr
#define TRACE(x...)
#endif

        .section .entry.text,"ax",%progbits
        .align 5
#ifdef !(IS_ENABLED(CONFIG_TRACE_IRQFLAGS) || IS_ENABLED(CONFIG_CONTEXT_TRACKING) || \
        IS_ENABLED(CONFIG_DEBUG_RSEQ))
/*
 * This is the fast syscall return path. We do as little as possible here,
 * such as avoiding writing r0 to the stack. We only use this path if we
 * have tracing, context tracking and rseq debug disabled - the overheads

```

```

percy@percy-P65-67R5RP:~/kernel/linux-5.3.1$ make ARCH=arm CROSS_COMPILE=arm-linux-gnueabi- defconfig
make[1]: Warning: File 'include/config/auto.conf' has modification time 23515 s in the future
make[1]: warning: Clock skew detected. Your build may be incomplete.
make[1]: Warning: File 'include/config/auto.conf' has modification time 23515 s in the future
*** Default configuration is based on 'multi_v7_defconfig'
#
# No change to .config
#
make[1]: warning: Clock skew detected. Your build may be incomplete.
percy@percy-P65-67R5RP:~/kernel/linux-5.3.1$ make ARCH=arm CROSS_COMPILE=arm-linux-gnueabi- arch/arm/kernel/entry-common.s
make: Warning: File 'include/config/auto.conf.cmd' has modification time 23505 s in the future
make[1]: Warning: File 'include/config/auto.conf' has modification time 23505 s in the future
make[1]: warning: Clock skew detected. Your build may be incomplete.
make[1]: Warning: File 'include/config/auto.conf' has modification time 23505 s in the future
make[1]: warning: Clock skew detected. Your build may be incomplete.
make[1]: Warning: File 'include/config/auto.conf' has modification time 23505 s in the future
make[1]: warning: Clock skew detected. Your build may be incomplete.
make[1]: Warning: File 'include/config/auto.conf' has modification time 23505 s in the future
make[1]: warning: Clock skew detected. Your build may be incomplete.
make[1]: Warning: File 'include/config/auto.conf' has modification time 23505 s in the future
make[1]: warning: Clock skew detected. Your build may be incomplete.
make[1]: Warning: File 'scripts/mod/.sumversion.o.cmd' has modification time 23506 s in the future
make[1]: warning: Clock skew detected. Your build may be incomplete.
make[1]: Warning: File 'kernel/.bounds.s.cmd' has modification time 23506 s in the future
CALL scripts/checksyscalls.sh
CALL scripts/atomic/check-atomics.sh
make[1]: warning: Clock skew detected. Your build may be incomplete.
make[1]: Warning: File 'include/config/auto.conf' has modification time 23504 s in the future
CPP arch/arm/kernel/entry-common.s
make[1]: warning: Clock skew detected. Your build may be incomplete.
make: warning: Clock skew detected. Your build may be incomplete.
percy@percy-P65-67R5RP:~/kernel/linux-5.3.1$

```

the whole system call table:

```
percy@percy-P65-67R5RP:~/kernel/linux-5.3.1/arch/arm/kernel$ cat entry-common.s | grep ", sys"
    adr tbl, sys_call_table @ load syscall table pointer
syscall 0, sys_restart_syscall
syscall 1, sys_exit
syscall 2, sys_fork
syscall 3, sys_read
syscall 4, sys_write
syscall 5, sys_open
syscall 6, sys_close
syscall 8, sys_creat
syscall 9, sys_link
syscall 10, sys_unlink
syscall 11, sys_execve
syscall 12, sys_chdir
syscall 14, sys_mknod
syscall 15, sys_chmod
syscall 16, sys_lchown16
syscall 19, sys_lseek
syscall 20, sys_getpid
syscall 21, sys_mount
syscall 23, sys_setuid16
syscall 24, sys_getuid16
syscall 26, sys_ptrace
syscall 29, sys_pause
syscall 33, sys_access
syscall 34, sys_nice
syscall 36, sys_sync
syscall 37, sys_kill
syscall 38, sys_rename
syscall 39, sys_mkdir
syscall 40, sys_rmdir
syscall 41, sys_dup
syscall 42, sys_pipe
syscall 43, sys_times
syscall 45, sys_brk
syscall 46, sys_setgid16
syscall 47, sys_getgid16
syscall 49, sys_geteuid16
syscall 50, sys_getegid16
syscall 51, sys_acct
syscall 52, sys_umount
syscall 54, sys_ioctl
syscall 55, sys_fcntl
syscall 57, sys_setpgid
```

```
syscall 60, sys_umask
syscall 61, sys_chroot
syscall 62, sys_ustat
syscall 63, sys_dup2
syscall 64, sys_getppid
syscall 65, sys_getpgrp
syscall 66, sys_setsid
syscall 67, sys_sigaction
syscall 70, sys_setreuid16
syscall 71, sys_setregid16
syscall 72, sys_sigsuspend
syscall 73, sys_sigpending
syscall 74, sys_sethostname
syscall 75, sys_setrlimit
syscall 77, sys_getrusage
syscall 78, sys_gettimeofday
syscall 79, sys_settimeofday
syscall 80, sys_getgroups16
syscall 81, sys_setgroups16
syscall 83, sys_symlink
syscall 85, sys_readlink
syscall 86, sys_uselib
syscall 87, sys_swapon
syscall 88, sys_reboot
syscall 91, sys_munmap
syscall 92, sys_truncate
syscall 93, sys_ftruncate
syscall 94, sys_fchmod
syscall 95, sys_fchown16
syscall 96, sys_getpriority
syscall 97, sys_setpriority
syscall 99, sys_statfs
syscall 100, sys_fstatfs
syscall 103, sys_syslog
syscall 104, sys_setitimer
syscall 105, sys_getitimer
syscall 106, sys_newstat
syscall 107, sys_newlstat
syscall 108, sys_newfstat
syscall 111, sys_vhangup
syscall 114, sys_wait4
syscall 115, sys_swapoff
syscall 116, sys_sysinfo
syscall 118, sys_fsync
syscall 119, sys_sigreturn_wrapper
```

```
syscall 120, sys_clone
syscall 121, sys_setdomainname
syscall 122, sys_newuname
syscall 124, sys_adjtimex_time32
syscall 125, sys_mprotect
syscall 126, sys_sigprocmask
syscall 128, sys_init_module
syscall 129, sys_delete_module
syscall 131, sys_quotactl
syscall 132, sys_getpgid
syscall 133, sys_fchdir
syscall 134, sys_bdflush
syscall 135, sys_sysfs
syscall 136, sys_personality
syscall 138, sys_setfsuid16
syscall 139, sys_setfsgid16
syscall 140, sys_llseek
syscall 141, sys_getdents
syscall 142, sys_select
syscall 143, sys_flock
syscall 144, sys_msync
syscall 145, sys_readv
syscall 146, sys_writev
syscall 147, sys_getsid
syscall 148, sys_fdatasync
syscall 149, sys_sysctl
syscall 150, sys_mlock
syscall 151, sys_munlock
syscall 152, sys_mlockall
syscall 153, sys_munlockall
syscall 154, sys_sched_setparam
syscall 155, sys_sched_getparam
syscall 156, sys_sched_setscheduler
syscall 157, sys_sched_getscheduler
syscall 158, sys_sched_yield
syscall 159, sys_sched_get_priority_max
syscall 160, sys_sched_get_priority_min
syscall 161, sys_sched_rr_get_interval_time32
syscall 162, sys_nanosleep_time32
syscall 163, sys_mremap
syscall 164, sys_setresuid16
syscall 165, sys_getresuid16
syscall 168, sys_poll
syscall 170, sys_setresgid16
syscall 171, sys_getresgid16
```

```
syscall 172, sys_prctl
syscall 173, sys_rt_sigreturn_wrapper
syscall 174, sys_rt_sigaction
syscall 175, sys_rt_sigprocmask
syscall 176, sys_rt_sigpending
syscall 177, sys_rt_sigtimedwait_time32
syscall 178, sys_rt_sigqueueinfo
syscall 179, sys_rt_sigsuspend
syscall 180, sys_pread64
syscall 181, sys_pwrite64
syscall 182, sys_chown16
syscall 183, sys_getcwd
syscall 184, sys_capget
syscall 185, sys_capset
syscall 186, sys_sigaltstack
syscall 187, sys_sendfile
syscall 190, sys_vfork
syscall 191, sys_getrlimit
syscall 192, sys_mmap2
syscall 193, sys_truncate64
syscall 194, sys_ftruncate64
syscall 195, sys_stat64
syscall 196, sys_lstat64
syscall 197, sys_fstat64
syscall 198, sys_lchown
syscall 199, sys_getuid
syscall 200, sys_getgid
syscall 201, sys_geteuid
syscall 202, sys_getegid
syscall 203, sys_setreuid
syscall 204, sys_setregid
syscall 205, sys_getgroups
syscall 206, sys_setgroups
syscall 207, sys_fchown
syscall 208, sys_setresuid
syscall 209, sys_getresuid
syscall 210, sys_setresgid
syscall 211, sys_getresgid
syscall 212, sys_chown
syscall 213, sys_setuid
syscall 214, sys_setgid
syscall 215, sys_setfsuid
syscall 216, sys_setfsgid
syscall 217, sys_getdents64
syscall 218, sys_pivot_root
```

```
syscall 219, sys_mincore
syscall 220, sys_madvise
syscall 221, sys_fcntl64
syscall 224, sys_gettid
syscall 225, sys_readahead
syscall 226, sys_setxattr
syscall 227, sys_lsetxattr
syscall 228, sys_fsetxattr
syscall 229, sys_getxattr
syscall 230, sys_lgetxattr
syscall 231, sys_fgetxattr
syscall 232, sys_listxattr
syscall 233, sys_llistxattr
syscall 234, sys_flistxattr
syscall 235, sys_removexattr
syscall 236, sys_lremovexattr
syscall 237, sys_fremovexattr
syscall 238, sys_tkill
syscall 239, sys_sendfile64
syscall 240, sys_futex_time32
syscall 241, sys_sched_setaffinity
syscall 242, sys_sched_getaffinity
syscall 243, sys_io_setup
syscall 244, sys_io_destroy
syscall 245, sys_io_getevents_time32
syscall 246, sys_io_submit
syscall 247, sys_io_cancel
syscall 248, sys_exit_group
syscall 249, sys_lookup_dcookie
syscall 250, sys_epoll_create
syscall 251, sys_epoll_ctl
syscall 252, sys_epoll_wait
syscall 253, sys_remap_file_pages
syscall 256, sys_set_tid_address
syscall 257, sys_timer_create
syscall 258, sys_timer_settime32
syscall 259, sys_timer_gettime32
syscall 260, sys_timer_getoverrun
syscall 261, sys_timer_delete
syscall 262, sys_clock_settime32
syscall 263, sys_clock_gettime32
syscall 264, sys_clock_getres_time32
syscall 265, sys_clock_nanosleep_time32
syscall 266, sys_statfs64_wrapper
syscall 267, sys_fstatfs64_wrapper
```

```
syscall 268, sys_tgkill
syscall 269, sys_utimes_time32
syscall 270, sys_arm_fadvise64_64
syscall 271, sys_pciconfig_iobase
syscall 272, sys_pciconfig_read
syscall 273, sys_pciconfig_write
syscall 274, sys_mq_open
syscall 275, sys_mq_unlink
syscall 276, sys_mq_timedsend_time32
syscall 277, sys_mq_timedreceive_time32
syscall 278, sys_mq_notify
syscall 279, sys_mq_getsetattr
syscall 280, sys_waitid
syscall 281, sys_socket
syscall 282, sys_bind
syscall 283, sys_connect
syscall 284, sys_listen
syscall 285, sys_accept
syscall 286, sys_getsockname
syscall 287, sys_getpeername
syscall 288, sys_socketpair
syscall 289, sys_send
syscall 290, sys_sendto
syscall 291, sys_recv
syscall 292, sys_recvfrom
syscall 293, sys_shutdown
syscall 294, sys_setsockopt
syscall 295, sys_getsockopt
syscall 296, sys_sendmsg
syscall 297, sys_recvmsg
syscall 298, sys_semop
syscall 299, sys_semget
syscall 300, sys_old_semctl
syscall 301, sys_msgsnd
syscall 302, sys_msgrcv
syscall 303, sys_msgget
syscall 304, sys_old_msgctl
syscall 305, sys_shmat
syscall 306, sys_shmdt
syscall 307, sys_shmget
syscall 308, sys_old_shmctl
syscall 309, sys_add_key
syscall 310, sys_request_key
syscall 311, sys_keyctl
syscall 312, sys_senedimedop_time32
```

```
syscall 314, sys_ioprio_set
syscall 315, sys_ioprio_get
syscall 316, sys_inotify_init
syscall 317, sys_inotify_add_watch
syscall 318, sys_inotify_rm_watch
syscall 319, sys_mbind
syscall 320, sys_get_mempolicy
syscall 321, sys_set_mempolicy
syscall 322, sys_openat
syscall 323, sys_mkdirat
syscall 324, sys_mknodat
syscall 325, sys_fchownat
syscall 326, sys_futimesat_time32
syscall 327, sys_fstatat64
syscall 328, sys_unlinkat
syscall 329, sys_renameat
syscall 330, sys_linkat
syscall 331, sys_symlinkat
syscall 332, sys_readlinkat
syscall 333, sys_fchmodat
syscall 334, sys_faccessat
syscall 335, sys_pselect6_time32
syscall 336, sys_ppoll_time32
syscall 337, sys_unshare
syscall 338, sys_set_robust_list
syscall 339, sys_get_robust_list
syscall 340, sys_splice
syscall 341, sys_sync_file_range2
syscall 342, sys_tee
syscall 343, sys_vmsplice
syscall 344, sys_move_pages
syscall 345, sys_getcpu
syscall 346, sys_epoll_pwait
syscall 347, sys_kexec_load
syscall 348, sys_utimensat_time32
syscall 349, sys_signalfd
syscall 350, sys_timerfd_create
syscall 351, sys_eventfd
syscall 352, sys_fallocate
syscall 353, sys_timerfd_settime32
syscall 354, sys_timerfd_gettime32
syscall 355, sys_signalfd4
syscall 356, sys_eventfd2
syscall 357, sys_epoll_create1
syscall 358, sys_dup3
```

```
syscall 359, sys_pipe2
syscall 360, sys_inotify_init1
syscall 361, sys_preadv
syscall 362, sys_pwritev
syscall 363, sys_rt_tgsigqueueinfo
syscall 364, sys_perf_event_open
syscall 365, sys_recvmmsg_time32
syscall 366, sys_accept4
syscall 367, sys_fanotify_init
syscall 368, sys_fanotify_mark
syscall 369, sys_prlimit64
syscall 370, sys_name_to_handle_at
syscall 371, sys_open_by_handle_at
syscall 372, sys_clock_adjtime32
syscall 373, sys_syncfs
syscall 374, sys_sendmmsg
syscall 375, sys_setns
syscall 376, sys_process_vm_readv
syscall 377, sys_process_vm_writev
syscall 378, sys_kcmp
syscall 379, sys_finit_module
syscall 380, sys_sched_setattr
syscall 381, sys_sched_getattr
syscall 382, sys_renameat2
syscall 383, sys_seccomp
syscall 384, sys_getrandom
syscall 385, sys_memfd_create
syscall 386, sys_bpf
syscall 387, sys_execveat
syscall 388, sys_userfaultfd
syscall 389, sys_membarrier
syscall 390, sys_mlock2
syscall 391, sys_copy_file_range
syscall 392, sys_preadv2
syscall 393, sys_pwritev2
syscall 394, sys_pkey_mprotect
syscall 395, sys_pkey_alloc
syscall 396, sys_pkey_free
syscall 397, sys_statx
syscall 398, sys_rseq
syscall 399, sys_io_pgetevents_time32
syscall 400, sys_migrate_pages
syscall 401, sys_kexec_file_load
syscall 403, sys_clock_gettime
syscall 404, sys_clock_settime
```

```

syscall 405, sys_clock_adjtime
syscall 406, sys_clock_getres
syscall 407, sys_clock_nanosleep
syscall 408, sys_timer_gettime
syscall 409, sys_timer_settime
syscall 410, sys_timerfd_gettime
syscall 411, sys_timerfd_settime
syscall 412, sys_utimensat
syscall 413, sys_pselect6
syscall 414, sys_ppoll
syscall 416, sys_io_pgetevents
syscall 417, sys_recvmmsg
syscall 418, sys_mq_timedsend
syscall 419, sys_mq_timedreceive
syscall 420, sys_sentionedop
syscall 421, sys_rt_sigtimedwait
syscall 422, sys_futex
syscall 423, sys_sched_rr_get_interval
syscall 424, sys_pidfd_send_signal
syscall 425, sys_io_uring_setup
syscall 426, sys_io_uring_enter
syscall 427, sys_io_uring_register
syscall 428, sys_open_tree
syscall 429, sys_move_mount
syscall 430, sys_fsopen
syscall 431, sys_fsconfig
syscall 432, sys_fsmount
syscall 433, sys_fspick
syscall 434, sys_pidfd_open
syscall 435, sys_clone3

```

RISC-V(32 bit)

source code file: syscall_table.c → syscall_table.i

```

percy@percy-P65-67RSRP:~/kernel/linux-5.3.1/arch/riscv/kernel$ ls | grep "syscall_table.c"
syscall_table.c

```

```

percy@percy-P65-67RSRP:~/kernel/linux-5.3.1/arch/riscv/kernel$ cat syscall_table.c
// SPDX-License-Identifier: GPL-2.0-only
/*

```

```

 * Copyright (C) 2009 Arnd Bergmann <arnd@arndb.de>
 * Copyright (C) 2012 Regents of the University of California
 */

```

```

#include <linux/linkage.h>
#include <linux/syscalls.h>
#include <asm-generic/syscalls.h>
#include <asm/vdso.h>

```

```

#undef __SYSCALL
#define __SYSCALL(nr, call)    [nr] = (call),

void *sys_call_table[__NR_syscalls] = {
    [0 ... __NR_syscalls - 1] = sys_ni_syscall,
#include <asm/unistd.h>
};

```

```

percy@percy-P65-67RSRP:~/kernel/linux-5.3.1$ make ARCH=riscv CROSS_COMPILE=riscv/bin/riscv32-unknown-linux-gnu- defconfig
*** Default configuration is based on 'defconfig'
#
# No change to .config
#

```

```

percy@percy-P65-67RSRP:~/kernel/linux-5.3.1$ make ARCH=riscv CROSS_COMPILE=riscv/bin/riscv32-unknown-linux-gnu- arch/riscv/kernel/syscall_table.i
CALL scripts/checksyscalls.sh
CALL scripts/atomic/check-atomics.sh
percy@percy-P65-67RSRP:~/kernel/linux-5.3.1$

```

the whole system call table:

```
percy@percy-P65-67SRP:~/kernel/linux-5.3.1/arch/riscv/kernel$ cat syscall_table.i | grep "(sys_"
[0] = (sys_io_setup),
[1] = (sys_io_destroy),
[2] = (sys_io_submit),
[3] = (sys_io_cancel),
[4] = (sys_io_getevents),
[5] = (sys_setxattr),
[6] = (sys_lsetxattr),
[7] = (sys_fsetxattr),
[8] = (sys_getxattr),
[9] = (sys_lgetxattr),
[10] = (sys_fgetxattr),
[11] = (sys_listxattr),
[12] = (sys_llistxattr),
[13] = (sys_flistxattr),
[14] = (sys_removexattr),
[15] = (sys_lremovexattr),
[16] = (sys_fremovexattr),
[17] = (sys_getcwd),
[18] = (sys_lookup_dcookie),
[19] = (sys_eventfd2),
[20] = (sys_epoll_create1),
[21] = (sys_epoll_ctl),
[22] = (sys_epoll_pwait),
[23] = (sys_dup),
[24] = (sys_dup3),
[25] = (sys_fcntl),
[26] = (sys_inotify_init1),
[27] = (sys_inotify_add_watch),
[28] = (sys_inotify_rm_watch),
[29] = (sys_ioctl),
[30] = (sys_ioprio_set),
[31] = (sys_ioprio_get),
[32] = (sys_flock),
[33] = (sys_mknodat),
[34] = (sys_mkdirat),
[35] = (sys_unlinkat),
[36] = (sys_symlinkat),
[37] = (sys_linkat),
[39] = (sys_umount),
[40] = (sys_mount),
[41] = (sys_pivot_root),
[42] = (sys_ni_syscall),
[43] = (sys_statfs),
```

```
[44] = (sys_fstatfs),
[45] = (sys_truncate),
[46] = (sys_ftruncate),
[47] = (sys_fallocate),
[48] = (sys_faccessat),
[49] = (sys_chdir),
[50] = (sys_fchdir),
[51] = (sys_chroot),
[52] = (sys_fchmod),
[53] = (sys_fchmodat),
[54] = (sys_fchownat),
[55] = (sys_fchown),
[56] = (sys_openat),
[57] = (sys_close),
[58] = (sys_vhangup),
[59] = (sys_pipe2),
[60] = (sys_quotactl),
[61] = (sys_getdents64),
[62] = (sys_lseek),
[63] = (sys_read),
[64] = (sys_write),
[65] = (sys_readv),
[66] = (sys_writev),
[67] = (sys_pread64),
[68] = (sys_pwrite64),
[69] = (sys_preadv),
[70] = (sys_pwritev),
[71] = (sys_sendfile64),
[72] = (sys_pselect6),
[73] = (sys_ppoll),
[74] = (sys_signalfd4),
[75] = (sys_vmsplice),
[76] = (sys_splice),
[77] = (sys_tee),
[78] = (sys_readlinkat),
[79] = (sys_newfstatat),
[80] = (sys_newfstat),
[81] = (sys_sync),
[82] = (sys_fsync),
[83] = (sys_fdatasync),
[84] = (sys_sync_file_range),
[85] = (sys_timerfd_create),
[86] = (sys_timerfd_settime),
[87] = (sys_timerfd_gettime),
[88] = (sys_utimensat),
```

```
[89] = (sys_acct),
[90] = (sys_capget),
[91] = (sys_capset),
[92] = (sys_personality),
[93] = (sys_exit),
[94] = (sys_exit_group),
[95] = (sys_waitid),
[96] = (sys_set_tid_address),
[97] = (sys_unshare),
[98] = (sys_futex),
[99] = (sys_set_robust_list),
[100] = (sys_get_robust_list),
[101] = (sys_nanosleep),
[102] = (sys_getitimer),
[103] = (sys_setitimer),
[104] = (sys_kexec_load),
[105] = (sys_init_module),
[106] = (sys_delete_module),
[107] = (sys_timer_create),
[108] = (sys_timer_gettime),
[109] = (sys_timer_getoverrun),
[110] = (sys_timer_settime),
[111] = (sys_timer_delete),
[112] = (sys_clock_settime),
[113] = (sys_clock_gettime),
[114] = (sys_clock_getres),
[115] = (sys_clock_nanosleep),
[116] = (sys_syslog),
[117] = (sys_ptrace),
[118] = (sys_sched_setparam),
[119] = (sys_sched_setscheduler),
[120] = (sys_sched_getscheduler),
[121] = (sys_sched_getparam),
[122] = (sys_sched_setaffinity),
[123] = (sys_sched_getaffinity),
[124] = (sys_sched_yield),
[125] = (sys_sched_get_priority_max),
[126] = (sys_sched_get_priority_min),
[127] = (sys_sched_rr_get_interval),
[128] = (sys_restart_syscall),
[129] = (sys_kill),
[130] = (sys_tkill),
[131] = (sys_tgkill),
[132] = (sys_sigaltstack),
[133] = (sys_rt_sigsuspend),
```

```
[134] = (sys_rt_sigaction),
[135] = (sys_rt_sigprocmask),
[136] = (sys_rt_sigpending),
[137] = (sys_rt_sigtimedwait),
[138] = (sys_rt_sigqueueinfo),
[139] = (sys_rt_sigreturn),
[140] = (sys_setpriority),
[141] = (sys_getpriority),
[142] = (sys_reboot),
[143] = (sys_setregid),
[144] = (sys_setgid),
[145] = (sys_setreuid),
[146] = (sys_setuid),
[147] = (sys_setresuid),
[148] = (sys_getresuid),
[149] = (sys_setresgid),
[150] = (sys_getresgid),
[151] = (sys_setfsuid),
[152] = (sys_setfsgid),
[153] = (sys_times),
[154] = (sys_setpgid),
[155] = (sys_getpgid),
[156] = (sys_getsid),
[157] = (sys_setsid),
[158] = (sys_getgroups),
[159] = (sys_setgroups),
[160] = (sys_newuname),
[161] = (sys_sethostname),
[162] = (sys_setdomainname),
[163] = (sys_getrlimit),
[164] = (sys_setrlimit),
[165] = (sys_getrusage),
[166] = (sys_umask),
[167] = (sys_prctl),
[168] = (sys_getcpu),
[169] = (sys_gettimeofday),
[170] = (sys_settimeofday),
[171] = (sys_adjtimex),
[172] = (sys_getpid),
[173] = (sys_getppid),
[174] = (sys_getuid),
[175] = (sys_geteuid),
[176] = (sys_getgid),
[177] = (sys_getegid),
[178] = (sys_gettid),
```

```
[179] = (sys_sysinfo),
[180] = (sys_mq_open),
[181] = (sys_mq_unlink),
[182] = (sys_mq_timedsend),
[183] = (sys_mq_timedreceive),
[184] = (sys_mq_notify),
[185] = (sys_mq_getsetattr),
[186] = (sys_msgget),
[187] = (sys_msgctl),
[188] = (sys_msgrcv),
[189] = (sys_msgsnd),
[190] = (sys_semget),
[191] = (sys_semctl),
[192] = (sys_senedop),
[193] = (sys_semop),
[194] = (sys_shmget),
[195] = (sys_shmctl),
[196] = (sys_shmat),
[197] = (sys_shmdt),
[198] = (sys_socket),
[199] = (sys_socketpair),
[200] = (sys_bind),
[201] = (sys_listen),
[202] = (sys_accept),
[203] = (sys_connect),
[204] = (sys_getsockname),
[205] = (sys_getpeername),
[206] = (sys_sendto),
[207] = (sys_recvfrom),
[208] = (sys_setsockopt),
[209] = (sys_getsockopt),
[210] = (sys_shutdown),
[211] = (sys_sendmsg),
[212] = (sys_recvmsg),
[213] = (sys_readahead),
[214] = (sys_brk),
[215] = (sys_munmap),
[216] = (sys_mremap),
[217] = (sys_add_key),
[218] = (sys_request_key),
[219] = (sys_keyctl),
[220] = (sys_clone),
[221] = (sys_execve),
[222] = (sys_mmap),
[223] = (sys_fadvise64_64).
```

```
[224] = (sys_swapon),
[225] = (sys_swapoff),
[226] = (sys_mprotect),
[227] = (sys_msync),
[228] = (sys_mlock),
[229] = (sys_munlock),
[230] = (sys_mlockall),
[231] = (sys_munlockall),
[232] = (sys_mincore),
[233] = (sys_madvise),
[234] = (sys_remap_file_pages),
[235] = (sys_mbind),
[236] = (sys_get_mempolicy),
[237] = (sys_set_mempolicy),
[238] = (sys_migrate_pages),
[239] = (sys_move_pages),
[240] = (sys_rt_tgsigqueueinfo),
[241] = (sys_perf_event_open),
[242] = (sys_accept4),
[243] = (sys_recvmmsg),
[244] = (sys_wait4),
[245] = (sys_prlimit64),
[246] = (sys_fanotify_init),
[247] = (sys_fanotify_mark),
[248] = (sys_name_to_handle_at),
[249] = (sys_open_by_handle_at),
[250] = (sys_clock_adjtime),
[251] = (sys_syncfs),
[252] = (sys_setns),
[253] = (sys_sendmmsg),
[254] = (sys_process_vm_readv),
[255] = (sys_process_vm_writev),
[256] = (sys_kcmp),
[257] = (sys_finit_module),
[258] = (sys_sched_setattr),
[259] = (sys_sched_getattr),
[260] = (sys_renameat2),
[261] = (sys_seccomp),
[262] = (sys_getrandom),
[263] = (sys_memfd_create),
[264] = (sys_bpf),
[265] = (sys_execveat),
[266] = (sys_userfaultfd),
[267] = (sys_membARRIER),
[268] = (sys_mlock2).
```

```

[285] = (sys_copy_file_range),
[286] = (sys_preadv2),
[287] = (sys_pwritev2),
[288] = (sys_pkey_mprotect),
[289] = (sys_pkey_alloc),
[290] = (sys_pkey_free),
[291] = (sys_statx),
[292] = (sys_io_pgetevents),
[293] = (sys_rseq),
[294] = (sys_kexec_file_load),
[424] = (sys_pidfd_send_signal),
[425] = (sys_io_uring_setup),
[426] = (sys_io_uring_enter),
[427] = (sys_io_uring_register),
[428] = (sys_open_tree),
[429] = (sys_move_mount),
[430] = (sys_fsopen),
[431] = (sys_fsconfig),
[432] = (sys_fsmount),
[433] = (sys_fspick),
[434] = (sys_pidfd_open),
[435] = (sys_clone3),
[(244 + 15)] = (sys_riscv_flush_icache),

```

RISC-V(64 bit)

source code file: syscall_table.c → syscall_table.i

```

percy@percy-P65-67RSRP:~/kernel/linux-5.3.1/arch/riscv/kernel$ ls | grep "syscall_table.c"
syscall_table.c
percy@percy-P65-67RSRP:~/kernel/linux-5.3.1/arch/riscv/kernel$ cat syscall_table.c
// SPDX-License-Identifier: GPL-2.0-only
/*
 * Copyright (C) 2009 Arnd Bergmann <arnd@arndb.de>
 * Copyright (C) 2012 Regents of the University of California
 */

#include <linux/linkage.h>
#include <linux/syscalls.h>
#include <asm-generic/syscalls.h>
#include <asm/vdso.h>

#ifdef __SYSCALL
#define __SYSCALL(nr, call)    [nr] = (call),

void *sys_call_table[__NR_syscalls] = {
    [0 ... __NR_syscalls - 1] = sys_ni_syscall,
#include <asm/unistd.h>
};

```

```

percy@percy-P65-67RSRP:~/kernel/linux-5.3.1$ make ARCH=riscv CROSS_COMPILE=/home/percy/Downloads/riscv64-unknown-linux-gnu-gcc9.2/riscv64/bin/riscv64-unknown-linux-gnu- defconfig
*** Default configuration is based on 'defconfig'
#
# configuration written to .config
percy@percy-P65-67RSRP:~/kernel/linux-5.3.1$ make ARCH=riscv CROSS_COMPILE=/home/percy/Downloads/riscv64-unknown-linux-gnu-gcc9.2/riscv64/bin/riscv64-unknown-linux-gnu- arch/riscv/kernel/syscall_table.i
scripts/kconfig/conf  --syncconfig Kconfig
CC      scripts/mod/empty.o
MKELF   scripts/mod/elfconfig.h
HOSTCC  scripts/mod/modpost.o
CC      scripts/mod/devicetable-offsets.s
UPD      scripts/mod/devicetable-offsets.h
HOSTCC  scripts/mod/file2alias.o
HOSTCC  scripts/mod/juniverse.o
HOSTLD  scripts/mod/modpost
CC      kernel/bounds.s
UPD      include/generated/bounds.h
UPD      include/generated/lineconst.h
CC      arch/riscv/kernel/asm-offsets.s
UPD      include/generated/asm-offsets.h
CALL    scripts/checksyscalls.sh
CALL    scripts/atomic/check-atomics.sh
CPP      arch/riscv/kernel/syscall_table.i

```

the whole system call table:

```
percy@percy-P65-67RSRP:~/kernel/linux-5.3.1/arch/riscv/kernel$ cat syscall_table.i | grep "= (sys"
[0] = (sys_io_setup),
[1] = (sys_io_destroy),
[2] = (sys_io_submit),
[3] = (sys_io_cancel),
[4] = (sys_io_getevents),
[5] = (sys_setxattr),
[6] = (sys_lsetxattr),
[7] = (sys_fsetxattr),
[8] = (sys_getxattr),
[9] = (sys_lgetxattr),
[10] = (sys_fgetxattr),
[11] = (sys_listxattr),
[12] = (sys_llistxattr),
[13] = (sys_flistxattr),
[14] = (sys_removexattr),
[15] = (sys_lremovexattr),
[16] = (sys_fremovexattr),
[17] = (sys_getcwd),
[18] = (sys_lookup_dcookie),
[19] = (sys_eventfd2),
[20] = (sys_epoll_create1),
[21] = (sys_epoll_ctl),
[22] = (sys_epoll_pwait),
[23] = (sys_dup),
[24] = (sys_dup3),
[25] = (sys_fcntl),
[26] = (sys_inotify_init1),
[27] = (sys_inotify_add_watch),
[28] = (sys_inotify_rm_watch),
[29] = (sys_ioctl),
[30] = (sys_ioprio_set),
[31] = (sys_ioprio_get),
[32] = (sys_flock),
[33] = (sys_mknodat),
[34] = (sys_mkdirat),
[35] = (sys_unlinkat),
[36] = (sys_symlinkat),
[37] = (sys_linkat),
[39] = (sys_umount),
[40] = (sys_mount),
[41] = (sys_pivot_root),
[42] = (sys_ni_syscall),
[43] = (sys_statfs),
[44] = (sys_fstatfs),
[45] = (sys_truncate),
[46] = (sys_ftruncate),
[47] = (sys_fallocate),
[48] = (sys_faccessat),
[49] = (sys_chdir),
[50] = (sys_fchdir),
[51] = (sys_chroot),
[52] = (sys_fchmod),
[53] = (sys_fchmodat),
[54] = (sys_fchownat),
[55] = (sys_fchown),
[56] = (sys_openat),
[57] = (sys_close),
[58] = (sys_vhangup),
[59] = (sys_pipe2),
[60] = (sys_quotactl),
```

```
[61] = (sys_getdents64),
[62] = (sys_lseek),
[63] = (sys_read),
[64] = (sys_write),
[65] = (sys_readv),
[66] = (sys_writev),
[67] = (sys_pread64),
[68] = (sys_pwrite64),
[69] = (sys_preadv),
[70] = (sys_pwritev),
[71] = (sys_sendfile64),
[72] = (sys_pselect6),
[73] = (sys_ppoll),
[74] = (sys_signalfd4),
[75] = (sys_vmsplice),
[76] = (sys_splice),
[77] = (sys_tee),
[78] = (sys_readlinkat),
[79] = (sys_newfstatat),
[80] = (sys_newfstat),
[81] = (sys_sync),
[82] = (sys_fsync),
[83] = (sys_fdatasync),
[84] = (sys_sync_file_range),
[85] = (sys_timerfd_create),
[86] = (sys_timerfd_settime),
[87] = (sys_timerfd_gettime),
[88] = (sys_utimensat),
[89] = (sys_acct),
[90] = (sys_capget),
[91] = (sys_capset),
[92] = (sys_personality),
[93] = (sys_exit),
[94] = (sys_exit_group),
[95] = (sys_waitid),
[96] = (sys_set_tid_address),
[97] = (sys_unshare),
[98] = (sys_futex),
[99] = (sys_set_robust_list),
[100] = (sys_get_robust_list),
[101] = (sys_nanosleep),
[102] = (sys_getitimer),
[103] = (sys_setitimer),
[104] = (sys_kexec_load),
[105] = (sys_init_module),
[106] = (sys_delete_module),
[107] = (sys_timer_create),
[108] = (sys_timer_gettime),
[109] = (sys_timer_getoverrun),
[110] = (sys_timer_settime),
[111] = (sys_timer_delete),
[112] = (sys_clock_settime),
[113] = (sys_clock_gettime),
[114] = (sys_clock_getres),
[115] = (sys_clock_nanosleep),
[116] = (sys_syslog),
[117] = (sys_ptrace),
[118] = (sys_sched_setparam),
[119] = (sys_sched_setscheduler),
[120] = (sys_sched_getscheduler),
[121] = (sys_sched_getparam),
```

```
[122] = (sys_sched_setaffinity),
[123] = (sys_sched_getaffinity),
[124] = (sys_sched_yield),
[125] = (sys_sched_get_priority_max),
[126] = (sys_sched_get_priority_min),
[127] = (sys_sched_rr_get_interval),
[128] = (sys_restart_syscall),
[129] = (sys_kill),
[130] = (sys_tkill),
[131] = (sys_tgkill),
[132] = (sys_sigaltstack),
[133] = (sys_rt_sigsuspend),
[134] = (sys_rt_sigaction),
[135] = (sys_rt_sigprocmask),
[136] = (sys_rt_sigpending),
[137] = (sys_rt_sigtimedwait),
[138] = (sys_rt_sigqueueinfo),
[139] = (sys_rt_sigreturn),
[140] = (sys_setpriority),
[141] = (sys_getpriority),
[142] = (sys_reboot),
[143] = (sys_setregid),
[144] = (sys_setgid),
[145] = (sys_setreuid),
[146] = (sys_setuid),
[147] = (sys_setresuid),
[148] = (sys_getresuid),
[149] = (sys_setresgid),
[150] = (sys_getresgid),
[151] = (sys_setfsuid),
[152] = (sys_setfsgid),
[153] = (sys_times),
[154] = (sys_setpgid),
[155] = (sys_getpgid),
[156] = (sys_getsid),
[157] = (sys_setsid),
[158] = (sys_getgroups),
[159] = (sys_setgroups),
[160] = (sys_newuname),
[161] = (sys_sethostname),
[162] = (sys_setdomainname),
[163] = (sys_getrlimit),
[164] = (sys_setrlimit),
[165] = (sys_getrusage),
[166] = (sys_umask),
[167] = (sys_prctl),
[168] = (sys_getcpu),
[169] = (sys_gettimeofday),
[170] = (sys_settimeofday),
[171] = (sys_adjtimex),
[172] = (sys_getpid),
[173] = (sys_getppid),
[174] = (sys_getuid),
[175] = (sys_geteuid),
[176] = (sys_getgid),
[177] = (sys_getegid),
[178] = (sys_gettid),
[179] = (sys_sysinfo),
[180] = (sys_mq_open),
[181] = (sys_mq_unlink),
[182] = (sys_mq_timedsend),
```

```
[183] = (sys_mq_timedreceive),
[184] = (sys_mq_notify),
[185] = (sys_mq_getsetattr),
[186] = (sys_msgget),
[187] = (sys_msgctl),
[188] = (sys_msgrcv),
[189] = (sys_msgsnd),
[190] = (sys_semget),
[191] = (sys_semctl),
[192] = (sys_semtimedop),
[193] = (sys_semop),
[194] = (sys_shmget),
[195] = (sys_shmctl),
[196] = (sys_shmat),
[197] = (sys_shmdt),
[198] = (sys_socket),
[199] = (sys_socketpair),
[200] = (sys_bind),
[201] = (sys_listen),
[202] = (sys_accept),
[203] = (sys_connect),
[204] = (sys_getsockname),
[205] = (sys_getpeername),
[206] = (sys_sendto),
[207] = (sys_recvfrom),
[208] = (sys_setsockopt),
[209] = (sys_getsockopt),
[210] = (sys_shutdown),
[211] = (sys_sendmsg),
[212] = (sys_recvmsg),
[213] = (sys_readahead),
[214] = (sys_brk),
[215] = (sys_munmap),
[216] = (sys_mremap),
[217] = (sys_add_key),
[218] = (sys_request_key),
[219] = (sys_keyctl),
[220] = (sys_clone),
[221] = (sys_execve),
[222] = (sys_mmap),
[223] = (sys_fadvise64_64),
[224] = (sys_swapon),
[225] = (sys_swapoff),
[226] = (sys_mprotect),
[227] = (sys_msync),
[228] = (sys_mlock),
[229] = (sys_munlock),
[230] = (sys_mlockall),
[231] = (sys_munlockall),
[232] = (sys_mincore),
[233] = (sys_madvise),
[234] = (sys_remap_file_pages),
[235] = (sys_mbind),
[236] = (sys_get_mempolicy),
[237] = (sys_set_mempolicy),
[238] = (sys_migrate_pages),
[239] = (sys_move_pages),
[240] = (sys_rt_tgsigqueueinfo),
[241] = (sys_perf_event_open),
[242] = (sys_accept4),
[243] = (sys_recvmmsg),
```

```

[260] = (sys_wait4),
[261] = (sys_prlimit64),
[262] = (sys_fanotify_init),
[263] = (sys_fanotify_mark),
[264] = (sys_name_to_handle_at),
[265] = (sys_open_by_handle_at),
[266] = (sys_clock_adjtime),
[267] = (sys_syncfs),
[268] = (sys_setns),
[269] = (sys_sendmmsg),
[270] = (sys_process_vm_readv),
[271] = (sys_process_vm_writev),
[272] = (sys_kcmp),
[273] = (sys_finit_module),
[274] = (sys_sched_setattr),
[275] = (sys_sched_getattr),
[276] = (sys_renameat2),
[277] = (sys_seccomp),
[278] = (sys_getrandom),
[279] = (sys_memfd_create),
[280] = (sys_bpf),
[281] = (sys_execveat),
[282] = (sys_userfaultfd),
[283] = (sys_membARRIER),
[284] = (sys_mlock2),
[285] = (sys_copy_file_range),
[286] = (sys_preadv2),
[287] = (sys_pwritev2),
[288] = (sys_pkey_mprotect),
[289] = (sys_pkey_alloc),
[290] = (sys_pkey_free),
[291] = (sys_statx),
[292] = (sys_io_pgetevents),
[293] = (sys_rseq),
[294] = (sys_kexec_file_load),
[424] = (sys_pidfd_send_signal),
[425] = (sys_io_uring_setup),
[426] = (sys_io_uring_enter),
[427] = (sys_io_uring_register),
[428] = (sys_open_tree),
[429] = (sys_move_mount),
[430] = (sys_fsopen),
[431] = (sys_fsconfig),
[432] = (sys_fsmount),
[433] = (sys_fspick),
[434] = (sys_pidfd_open),
[435] = (sys_clone3),
[(244 + 15)] = (sys_riscv_flush_icache),

```

X86(32 bit)

source code file: syscall_32.i → syscall_32.i

```

percy@percy-P65-67R5RP:~/kernel/linux-5.3.1/arch/x86/entry$ ls | grep syscall_32.c
syscall_32.c
percy@percy-P65-67R5RP:~/kernel/linux-5.3.1/arch/x86/entry$ cat syscall_32.c
// SPDX-License-Identifier: GPL-2.0
/* System call table for i386. */

#include <linux/linkage.h>
#include <linux/sys.h>
#include <linux/cache.h>
#include <asm/asm-offsets.h>
#include <asm/syscall.h>

#ifdef CONFIG_IA32_EMULATION
/* On X86_64, we use struct pt_regs * to pass parameters to syscalls */
#define __SYSCALL_I386(nr, sym, qual) extern asmlinkage long sym(const struct pt_regs *);

/* this is a lie, but it does not hurt as sys_ni_syscall just returns -EINVAL */
extern asmlinkage long sys_ni_syscall(const struct pt_regs *);

#else /* CONFIG_IA32_EMULATION */
#define __SYSCALL_I386(nr, sym, qual) extern asmlinkage long sym(unsigned long, unsigned long, unsigned long, unsigned long, unsigned long, unsigned long);
extern asmlinkage long sys_ni_syscall(unsigned long, unsigned long, unsigned long, unsigned long, unsigned long, unsigned long);
#endif /* CONFIG_IA32_EMULATION */

#include <asm/syscalls_32.h>
#undef __SYSCALL_I386

#define __SYSCALL_I386(nr, sym, qual) [nr] = sym,

__visible const sys_call_ptr_t ia32_sys_call_table[__NR_syscall_compat_max+1] = {
    /*
     * Smells like a compiler bug -- it doesn't work
     * when the & below is removed.
     */
    [0 ... __NR_syscall_compat_max] = &sys_ni_syscall,
#include <asm/syscalls_32.h>
};

```

```
percy@percy-P65-67RSRP:~/kernel/linux-5.3.1$ make ARCH=x86 CROSS_COMPILE= defconfig
*** Default configuration is based on 'x86_64_defconfig'
#
# configuration written to .config
#
```

```
percy@percy-P65-67RSRP:~/kernel/linux-5.3.1$ make ARCH=x86 CROSS_COMPILE= arch/x86/entry/syscall_32.i
CALL      scripts/checksyscalls.sh
CALL      scripts/atomic/check-atomics.sh
DESCEND    objtool
HOSTCC     /home/percy/kernel/linux-5.3.1/tools/objtool/fixdep.o
HOSTLD     /home/percy/kernel/linux-5.3.1/tools/objtool/fixdep-in.o
LINK       /home/percy/kernel/linux-5.3.1/tools/objtool/fixdep
CC         /home/percy/kernel/linux-5.3.1/tools/objtool/exec-cmd.o
CC         /home/percy/kernel/linux-5.3.1/tools/objtool/help.o
CC         /home/percy/kernel/linux-5.3.1/tools/objtool/pager.o
CC         /home/percy/kernel/linux-5.3.1/tools/objtool/parse-options.o
CC         /home/percy/kernel/linux-5.3.1/tools/objtool/run-command.o
CC         /home/percy/kernel/linux-5.3.1/tools/objtool/sigchain.o
CC         /home/percy/kernel/linux-5.3.1/tools/objtool/subcmd-config.o
LD         /home/percy/kernel/linux-5.3.1/tools/objtool/libsubcmd-in.o
AR         /home/percy/kernel/linux-5.3.1/tools/objtool/libsubcmd.a
GEN        /home/percy/kernel/linux-5.3.1/tools/objtool/arch/x86/lib/inat-tables.c
CC         /home/percy/kernel/linux-5.3.1/tools/objtool/arch/x86/decode.o
LD         /home/percy/kernel/linux-5.3.1/tools/objtool/arch/x86/objtool-in.o
CC         /home/percy/kernel/linux-5.3.1/tools/objtool/builtin-check.o
CC         /home/percy/kernel/linux-5.3.1/tools/objtool/builtin-orc.o
CC         /home/percy/kernel/linux-5.3.1/tools/objtool/check.o
CC         /home/percy/kernel/linux-5.3.1/tools/objtool/orc_gen.o
CC         /home/percy/kernel/linux-5.3.1/tools/objtool/orc_dump.o
CC         /home/percy/kernel/linux-5.3.1/tools/objtool/elf.o
CC         /home/percy/kernel/linux-5.3.1/tools/objtool/special.o
CC         /home/percy/kernel/linux-5.3.1/tools/objtool/objtool.o
CC         /home/percy/kernel/linux-5.3.1/tools/objtool/libstring.o
CC         /home/percy/kernel/linux-5.3.1/tools/objtool/libctype.o
CC         /home/percy/kernel/linux-5.3.1/tools/objtool/str_error_r.o
LD         /home/percy/kernel/linux-5.3.1/tools/objtool/objtool-in.o
LINK       /home/percy/kernel/linux-5.3.1/tools/objtool/objtool
CPP        arch/x86/entry/syscall_32.i
```

the whole system call table:

```
percy@percy-P65-67RSRP:~/kernel/linux-5.3.1/arch/x86/entry$ cat syscall_32.i | grep "=" | grep "sys_"
__attribute__((__externally_visible__)) const sys_call_ptr_t ia32_sys_call_table[435 + 1] = {
[0 ... 435] = &sys_ni_syscall,
[0] = __ia32_sys_restart_syscall,
[1] = __ia32_sys_exit,
[2] = __ia32_sys_fork,
[3] = __ia32_sys_read,
[4] = __ia32_sys_write,
[5] = __ia32_compat_sys_open,
[6] = __ia32_sys_close,
[7] = __ia32_sys_waitpid,
[8] = __ia32_sys_creat,
[9] = __ia32_sys_link,
[10] = __ia32_sys_unlink,
[11] = __ia32_compat_sys_execve,
[12] = __ia32_sys_chdir,
[13] = __ia32_sys_time32,
[14] = __ia32_sys_mknod,
[15] = __ia32_sys_chmod,
[16] = __ia32_sys_lchown16,
[18] = __ia32_sys_stat,
[19] = __ia32_compat_sys_lseek,
[20] = __ia32_sys_getpid,
[21] = __ia32_compat_sys_mount,
[22] = __ia32_sys_oldumount,
[23] = __ia32_sys_setuid16,
[24] = __ia32_sys_getuid16,
[25] = __ia32_sys_stime32,
[26] = __ia32_compat_sys_ptrace,
[27] = __ia32_sys_alarm,
[28] = __ia32_sys_fstat,
[29] = __ia32_sys_pause,
[30] = __ia32_sys_utime32,
[33] = __ia32_sys_access,
[34] = __ia32_sys_nice,
[36] = __ia32_sys_sync,
[37] = __ia32_sys_kill,
[38] = __ia32_sys_rename,
[39] = __ia32_sys_mkdir,
[40] = __ia32_sys_rmdir,
[41] = __ia32_sys_dup,
[42] = __ia32_sys_pipe,
[43] = __ia32_compat_sys_times,
[45] = __ia32_sys_brk,
[46] = __ia32_sys_setgid16,
```

```
[47] = __ia32_sys_getgid16,  
[48] = __ia32_sys_signal,  
[49] = __ia32_sys_geteuid16,  
[50] = __ia32_sys_getegid16,  
[51] = __ia32_sys_acct,  
[52] = __ia32_sys_umount,  
[54] = __ia32_compat_sys_ioctl,  
[55] = __ia32_compat_sys_fcntl64,  
[57] = __ia32_sys_setpgid,  
[59] = __ia32_sys_olduname,  
[60] = __ia32_sys_umask,  
[61] = __ia32_sys_chroot,  
[62] = __ia32_compat_sys_ustat,  
[63] = __ia32_sys_dup2,  
[64] = __ia32_sys_getppid,  
[65] = __ia32_sys_getpgrp,  
[66] = __ia32_sys_setsid,  
[67] = __ia32_compat_sys_sigaction,  
[68] = __ia32_sys_sgetmask,  
[69] = __ia32_sys_ssetmask,  
[70] = __ia32_sys_setreuid16,  
[71] = __ia32_sys_setregid16,  
[72] = __ia32_sys_sigsuspend,  
[73] = __ia32_compat_sys_sigpending,  
[74] = __ia32_sys_sethostname,  
[75] = __ia32_compat_sys_setrlimit,  
[76] = __ia32_compat_sys_old_getrlimit,  
[77] = __ia32_compat_sys_getrusage,  
[78] = __ia32_compat_sys_gettimeofday,  
[79] = __ia32_compat_sys_settimeofday,  
[80] = __ia32_sys_getgroups16,  
[81] = __ia32_sys_setgroups16,  
[82] = __ia32_compat_sys_old_select,  
[83] = __ia32_sys_symlink,  
[84] = __ia32_sys_lstat,  
[85] = __ia32_sys_readlink,  
[86] = __ia32_sys_uselib,  
[87] = __ia32_sys_swapon,  
[88] = __ia32_sys_reboot,  
[89] = __ia32_compat_sys_old_readdir,  
[90] = __ia32_compat_sys_x86_mmap,  
[91] = __ia32_sys_munmap,  
[92] = __ia32_compat_sys_truncate,  
[93] = __ia32_compat_sys_ftruncate,  
[94] = __ia32_sys_fchmod,
```

```
[95] = __ia32_sys_fchown16,  
[96] = __ia32_sys_getpriority,  
[97] = __ia32_sys_setpriority,  
[99] = __ia32_compat_sys_statfs,  
[100] = __ia32_compat_sys_fstatfs,  
[101] = __ia32_sys_ioperm,  
[102] = __ia32_compat_sys_socketcall,  
[103] = __ia32_sys_syslog,  
[104] = __ia32_compat_sys_setitimer,  
[105] = __ia32_compat_sys_getitimer,  
[106] = __ia32_compat_sys_newstat,  
[107] = __ia32_compat_sys_newlstat,  
[108] = __ia32_compat_sys_newfstat,  
[109] = __ia32_sys_uname,  
[110] = __ia32_sys_iopl,  
[111] = __ia32_sys_vhangup,  
[113] = sys_ni_syscall,  
[114] = __ia32_compat_sys_wait4,  
[115] = __ia32_sys_swapoff,  
[116] = __ia32_compat_sys_sysinfo,  
[117] = __ia32_compat_sys_ipc,  
[118] = __ia32_sys_fsync,  
[120] = __ia32_compat_sys_x86_clone,  
[121] = __ia32_sys_setdomainname,  
[122] = __ia32_sys_newuname,  
[123] = __ia32_sys_modify_ldt,  
[124] = __ia32_sys_adjtimex_time32,  
[125] = __ia32_sys_mprotect,  
[126] = __ia32_compat_sys_sigprocmask,  
[128] = __ia32_sys_init_module,  
[129] = __ia32_sys_delete_module,  
[131] = __ia32_compat_sys_quotactl32,  
[132] = __ia32_sys_setpgid,  
[133] = __ia32_sys_fchdir,  
[134] = __ia32_sys_bdflush,  
[135] = __ia32_sys_sysfs,  
[136] = __ia32_sys_personality,  
[138] = __ia32_sys_setfsuid16,  
[139] = __ia32_sys_setfsgid16,  
[140] = __ia32_sys_llseek,  
[141] = __ia32_compat_sys_getdents,  
[142] = __ia32_compat_sys_select,  
[143] = __ia32_sys_flock,  
[144] = __ia32_sys_msync,  
[145] = __ia32_compat_sys_readv,
```

```
[146] = __ia32_compat_sys_writev,  
[147] = __ia32_sys_getsid,  
[148] = __ia32_sys_fdatasync,  
[149] = __ia32_compat_sys_sysctl,  
[150] = __ia32_sys_mlock,  
[151] = __ia32_sys_munlock,  
[152] = __ia32_sys_mlockall,  
[153] = __ia32_sys_munlockall,  
[154] = __ia32_sys_sched_setparam,  
[155] = __ia32_sys_sched_getparam,  
[156] = __ia32_sys_sched_setscheduler,  
[157] = __ia32_sys_sched_getscheduler,  
[158] = __ia32_sys_sched_yield,  
[159] = __ia32_sys_sched_get_priority_max,  
[160] = __ia32_sys_sched_get_priority_min,  
[161] = __ia32_sys_sched_rr_get_interval_time32,  
[162] = __ia32_sys_nanosleep_time32,  
[163] = __ia32_sys_mremap,  
[164] = __ia32_sys_setresuid16,  
[165] = __ia32_sys_getresuid16,  
[166] = sys_ni_syscall,  
[168] = __ia32_sys_poll,  
[170] = __ia32_sys_setresgid16,  
[171] = __ia32_sys_getresgid16,  
[172] = __ia32_sys_prctl,  
[174] = __ia32_compat_sys_rt_sigaction,  
[175] = __ia32_sys_rt_sigprocmask,  
[176] = __ia32_compat_sys_rt_sigpending,  
[177] = __ia32_compat_sys_rt_sigtimedwait_time32,  
[178] = __ia32_compat_sys_rt_sigqueueinfo,  
[179] = __ia32_sys_rt_sigsuspend,  
[180] = __ia32_compat_sys_x86_pread,  
[181] = __ia32_compat_sys_x86_pwrite,  
[182] = __ia32_sys_chown16,  
[183] = __ia32_sys_getcwd,  
[184] = __ia32_sys_capget,  
[185] = __ia32_sys_capset,  
[186] = __ia32_compat_sys_sigaltstack,  
[187] = __ia32_compat_sys_sendfile,  
[190] = __ia32_sys_vfork,  
[191] = __ia32_compat_sys_getrlimit,  
[192] = __ia32_sys_mmap_pgoff,  
[193] = __ia32_compat_sys_x86_truncate64,  
[194] = __ia32_compat_sys_x86_ftruncate64,  
[195] = __ia32_compat_sys_x86_stat64,
```

```
[196] = __ia32_compat_sys_x86_lstat64,  
[197] = __ia32_compat_sys_x86_fstat64,  
[198] = __ia32_sys_lchown,  
[199] = __ia32_sys_getuid,  
[200] = __ia32_sys_getgid,  
[201] = __ia32_sys_geteuid,  
[202] = __ia32_sys_getegid,  
[203] = __ia32_sys_setreuid,  
[204] = __ia32_sys_setregid,  
[205] = __ia32_sys_getgroups,  
[206] = __ia32_sys_setgroups,  
[207] = __ia32_sys_fchown,  
[208] = __ia32_sys_setresuid,  
[209] = __ia32_sys_getresuid,  
[210] = __ia32_sys_setresgid,  
[211] = __ia32_sys_getresgid,  
[212] = __ia32_sys_chown,  
[213] = __ia32_sys_setuid,  
[214] = __ia32_sys_setgid,  
[215] = __ia32_sys_setfsuid,  
[216] = __ia32_sys_setfsgid,  
[217] = __ia32_sys_pivot_root,  
[218] = __ia32_sys_mincore,  
[219] = __ia32_sys_madvise,  
[220] = __ia32_sys_getdents64,  
[221] = __ia32_compat_sys_fcntl64,  
[224] = __ia32_sys_gettid,  
[225] = __ia32_compat_sys_x86_readahead,  
[226] = __ia32_sys_setxattr,  
[227] = __ia32_sys_lsetxattr,  
[228] = __ia32_sys_fsetxattr,  
[229] = __ia32_sys_getxattr,  
[230] = __ia32_sys_lgetxattr,  
[231] = __ia32_sys_fgetxattr,  
[232] = __ia32_sys_listxattr,  
[233] = __ia32_sys_llistxattr,  
[234] = __ia32_sys_flistxattr,  
[235] = __ia32_sys_removexattr,  
[236] = __ia32_sys_lremovexattr,  
[237] = __ia32_sys_fremovexattr,  
[238] = __ia32_sys_tkill,  
[239] = __ia32_sys_sendfile64,  
[240] = __ia32_sys_futex_time32,  
[241] = __ia32_compat_sys_sched_setaffinity,  
[242] = __ia32_compat_sys_sched_getaffinity,
```

```
[243] = __ia32_sys_set_thread_area,  
[244] = __ia32_sys_get_thread_area,  
[245] = __ia32_compat_sys_io_setup,  
[246] = __ia32_sys_io_destroy,  
[247] = __ia32_sys_io_getevents_time32,  
[248] = __ia32_compat_sys_io_submit,  
[249] = __ia32_sys_io_cancel,  
[250] = __ia32_compat_sys_x86_fadvise64,  
[252] = __ia32_sys_exit_group,  
[253] = __ia32_compat_sys_lookup_dcookie,  
[254] = __ia32_sys_epoll_create,  
[255] = __ia32_sys_epoll_ctl,  
[256] = __ia32_sys_epoll_wait,  
[257] = __ia32_sys_remap_file_pages,  
[258] = __ia32_sys_set_tid_address,  
[259] = __ia32_compat_sys_timer_create,  
[260] = __ia32_sys_timer_settime32,  
[261] = __ia32_sys_timer_gettime32,  
[262] = __ia32_sys_timer_getoverrun,  
[263] = __ia32_sys_timer_delete,  
[264] = __ia32_sys_clock_settime32,  
[265] = __ia32_sys_clock_gettime32,  
[266] = __ia32_sys_clock_getres_time32,  
[267] = __ia32_sys_clock_nanosleep_time32,  
[268] = __ia32_compat_sys_statfs64,  
[269] = __ia32_compat_sys_fstatfs64,  
[270] = __ia32_sys_tgkill,  
[271] = __ia32_sys_utimes_time32,  
[272] = __ia32_compat_sys_x86_fadvise64_64,  
[274] = __ia32_sys_mbind,  
[275] = __ia32_compat_sys_get_mempolicy,  
[276] = __ia32_sys_set_mempolicy,  
[277] = __ia32_compat_sys_mq_open,  
[278] = __ia32_sys_mq_unlink,  
[279] = __ia32_sys_mq_timedsend_time32,  
[280] = __ia32_sys_mq_timedreceive_time32,  
[281] = __ia32_compat_sys_mq_notify,  
[282] = __ia32_compat_sys_mq_getsetattr,  
[283] = __ia32_compat_sys_kexec_load,  
[284] = __ia32_compat_sys_waitid,  
[286] = __ia32_sys_add_key,  
[287] = __ia32_sys_request_key,  
[288] = __ia32_compat_sys_keyctl,  
[289] = __ia32_sys_ioprio_set,  
[290] = __ia32_sys_ioprio_get,
```

```
[291] = __ia32_sys_inotify_init,  
[292] = __ia32_sys_inotify_add_watch,  
[293] = __ia32_sys_inotify_rm_watch,  
[294] = __ia32_sys_migrate_pages,  
[295] = __ia32_compat_sys_openat,  
[296] = __ia32_sys_mkdirat,  
[297] = __ia32_sys_mknodat,  
[298] = __ia32_sys_fchownat,  
[299] = __ia32_sys_futimesat_time32,  
[300] = __ia32_compat_sys_x86_fstatat,  
[301] = __ia32_sys_unlinkat,  
[302] = __ia32_sys_renameat,  
[303] = __ia32_sys_linkat,  
[304] = __ia32_sys_symlinkat,  
[305] = __ia32_sys_readlinkat,  
[306] = __ia32_sys_fchmodat,  
[307] = __ia32_sys_faccessat,  
[308] = __ia32_compat_sys_pselect6_time32,  
[309] = __ia32_compat_sys_ppoll_time32,  
[310] = __ia32_sys_unshare,  
[311] = __ia32_compat_sys_set_robust_list,  
[312] = __ia32_compat_sys_get_robust_list,  
[313] = __ia32_sys_splice,  
[314] = __ia32_compat_sys_x86_sync_file_range,  
[315] = __ia32_sys_tee,  
[316] = __ia32_compat_sys_vmsplice,  
[317] = __ia32_compat_sys_move_pages,  
[318] = __ia32_sys_getcpu,  
[319] = __ia32_sys_epoll_pwait,  
[320] = __ia32_sys_utimensat_time32,  
[321] = __ia32_compat_sys_signalfd,  
[322] = __ia32_sys_timerfd_create,  
[323] = __ia32_sys_eventfd,  
[324] = __ia32_compat_sys_x86_fallocate,  
[325] = __ia32_sys_timerfd_settime32,  
[326] = __ia32_sys_timerfd_gettime32,  
[327] = __ia32_compat_sys_signalfd4,  
[328] = __ia32_sys_eventfd2,  
[329] = __ia32_sys_epoll_create1,  
[330] = __ia32_sys_dup3,  
[331] = __ia32_sys_pipe2,  
[332] = __ia32_sys_inotify_init1,  
[333] = __ia32_compat_sys_preadv,  
[334] = __ia32_compat_sys_pwritev,  
[335] = __ia32_compat_sys_rt_tgsigqueueinfo,
```

```
[336] = __ia32_sys_perf_event_open,  
[337] = __ia32_compat_sys_recvmmsg_time32,  
[338] = __ia32_sys_fanotify_init,  
[339] = __ia32_compat_sys_fanotify_mark,  
[340] = __ia32_sys_prlimit64,  
[341] = __ia32_sys_name_to_handle_at,  
[342] = __ia32_compat_sys_open_by_handle_at,  
[343] = __ia32_sys_clock_adjtime32,  
[344] = __ia32_sys_syncfs,  
[345] = __ia32_compat_sys_sendmmsg,  
[346] = __ia32_sys_setns,  
[347] = __ia32_compat_sys_process_vm_readv,  
[348] = __ia32_compat_sys_process_vm_writev,  
[349] = __ia32_sys_kcmp,  
[350] = __ia32_sys_finit_module,  
[351] = __ia32_sys_sched_setattr,  
[352] = __ia32_sys_sched_getattr,  
[353] = __ia32_sys_renameat2,  
[354] = __ia32_sys_seccomp,  
[355] = __ia32_sys_getrandom,  
[356] = __ia32_sys_memfd_create,  
[357] = __ia32_sys_bpf,  
[358] = __ia32_compat_sys_execveat,  
[359] = __ia32_sys_socket,  
[360] = __ia32_sys_socketpair,  
[361] = __ia32_sys_bind,  
[362] = __ia32_sys_connect,  
[363] = __ia32_sys_listen,  
[364] = __ia32_sys_accept4,  
[365] = __ia32_compat_sys_getsockopt,  
[366] = __ia32_compat_sys_setsockopt,  
[367] = __ia32_sys_getsockname,  
[368] = __ia32_sys_getpeername,  
[369] = __ia32_sys_sendto,  
[370] = __ia32_compat_sys_sendmsg,  
[371] = __ia32_compat_sys_recvfrom,  
[372] = __ia32_compat_sys_recvmmsg,  
[373] = __ia32_sys_shutdown,  
[374] = __ia32_sys_userfaultfd,  
[375] = __ia32_sys_membARRIER,  
[376] = __ia32_sys_mlock2,  
[377] = __ia32_sys_copy_file_range,  
[378] = __ia32_compat_sys_preadv2,  
[379] = __ia32_compat_sys_pwritev2,  
[380] = __ia32_sys_pkey_mprotect,
```

```
[381] = __ia32_sys_pkey_alloc,  
[382] = __ia32_sys_pkey_free,  
[383] = __ia32_sys_statx,  
[384] = __ia32_compat_sys_arch_prctl,  
[385] = __ia32_compat_sys_io_pgetevents,  
[386] = __ia32_sys_rseq,  
[393] = __ia32_sys_semget,  
[394] = __ia32_compat_sys_semctl,  
[395] = __ia32_sys_shmget,  
[396] = __ia32_compat_sys_shmctl,  
[397] = __ia32_compat_sys_shmat,  
[398] = __ia32_sys_shmdt,  
[399] = __ia32_sys_msgget,  
[400] = __ia32_compat_sys_msgsnd,  
[401] = __ia32_compat_sys_msgrcv,  
[402] = __ia32_compat_sys_msgctl,  
[403] = __ia32_sys_clock_gettime,  
[404] = __ia32_sys_clock_settime,  
[405] = __ia32_sys_clock_adjtime,  
[406] = __ia32_sys_clock_getres,  
[407] = __ia32_sys_clock_nanosleep,  
[408] = __ia32_sys_timer_gettime,  
[409] = __ia32_sys_timer_settime,  
[410] = __ia32_sys_timerfd_gettime,  
[411] = __ia32_sys_timerfd_settime,  
[412] = __ia32_sys_utimensat,  
[413] = __ia32_compat_sys_pselect6_time64,  
[414] = __ia32_compat_sys_ppoll_time64,  
[416] = __ia32_sys_io_pgetevents,  
[417] = __ia32_compat_sys_recvmmsg_time64,  
[418] = __ia32_sys_mq_timedsend,  
[419] = __ia32_sys_mq_timedreceive,  
[420] = __ia32_sys_semtimedop,  
[421] = __ia32_compat_sys_rt_sigtimedwait_time64,  
[422] = __ia32_sys_futex,  
[423] = __ia32_sys_sched_rr_get_interval,  
[424] = __ia32_sys_pidfd_send_signal,  
[425] = __ia32_sys_io_uring_setup,  
[426] = __ia32_sys_io_uring_enter,  
[427] = __ia32_sys_io_uring_register,  
[428] = __ia32_sys_open_tree,  
[429] = __ia32_sys_move_mount,  
[430] = __ia32_sys_fsopen,  
[431] = __ia32_sys_fsconfig,  
[432] = __ia32_sys_fsmount,
```

```
[433] = __ia32_sys_fspick,  
[434] = __ia32_sys_pidfd_open,  
[435] = __ia32_sys_clone3,
```

X86_64

source code file: syscall_64.c → syscall_64.i

```
percy@percy-P65-67RSRP:~/kernel/linux-5.3.1/arch/x86/entry$ ls | grep syscall_64.c  
syscall_64.c  
percy@percy-P65-67RSRP:~/kernel/linux-5.3.1/arch/x86/entry$ cat syscall_64.c  
// SPDX-License-Identifier: GPL-2.0  
/* System call table for x86-64. */  
  
#include <linux/linkage.h>  
#include <linux/sys.h>  
#include <linux/cache.h>  
#include <asm/asm-offsets.h>  
#include <asm/syscall.h>  
  
/* this is a lie, but it does not hurt as sys_ni_syscall just returns -EINVAL */  
extern asmlinkage long sys_ni_syscall(const struct pt_regs *);  
#define __SYSCALL_64(nr, sym, qual) extern asmlinkage long sym(const struct pt_regs *);  
#include <asm/syscalls_64.h>  
#undef __SYSCALL_64  
  
#define __SYSCALL_64(nr, sym, qual) [nr] = sym,  
  
asmlinkage const sys_call_ptr_t sys_call_table[__NR_syscall_max+1] = {  
    /*  
     * Smells like a compiler bug -- it doesn't work  
     * when the & below is removed.  
     */  
    [0 ... __NR_syscall_max] = &sys_ni_syscall,  
#include <asm/syscalls_64.h>  
};
```

```
percy@percy-P65-67RSRP:~/kernel/linux-5.3.1$ make ARCH=x86 CROSS_COMPILE= defconfig  
*** Default configuration is based on 'x86_64_defconfig'  
#  
# No change to .config  
#  
percy@percy-P65-67RSRP:~/kernel/linux-5.3.1$ make ARCH=x86 CROSS_COMPILE= arch/x86/entry/syscall_64.i  
CALL scripts/checksyscalls.sh  
CALL scripts/atomic/check-atomics.sh  
DESCEND objtool  
CPP arch/x86/entry/syscall_64.i
```

the whole system call table:

```
percy@percy-P65-67R5RP:~/kernel/linux-5.3.1/arch/x86/entry$ cat syscall_64.i | grep "=" | grep "sys_"
const sys_call_ptr_t sys_call_table[435 +1] = {
[0 ... 435] = &sys_ni_syscall,
[0] = __x64_sys_read,
[1] = __x64_sys_write,
[2] = __x64_sys_open,
[3] = __x64_sys_close,
[4] = __x64_sys_newstat,
[5] = __x64_sys_newfstat,
[6] = __x64_sys_newlstat,
[7] = __x64_sys_poll,
[8] = __x64_sys_lseek,
[9] = __x64_sys_mmap,
[10] = __x64_sys_mprotect,
[11] = __x64_sys_munmap,
[12] = __x64_sys_brk,
[13] = __x64_sys_rt_sigaction,
[14] = __x64_sys_rt_sigprocmask,
[15] = __x64_sys_rt_sigreturn,
[16] = __x64_sys_ioctl,
[17] = __x64_sys_pread64,
[18] = __x64_sys_pwrite64,
[19] = __x64_sys_readv,
[20] = __x64_sys_writev,
[21] = __x64_sys_access,
[22] = __x64_sys_pipe,
[23] = __x64_sys_select,
[24] = __x64_sys_sched_yield,
[25] = __x64_sys_mremap,
[26] = __x64_sys_msync,
[27] = __x64_sys_mincore,
[28] = __x64_sys_madvise,
[29] = __x64_sys_shmget,
[30] = __x64_sys_shmat,
[31] = __x64_sys_shmctl,
[32] = __x64_sys_dup,
[33] = __x64_sys_dup2,
[34] = __x64_sys_pause,
[35] = __x64_sys_nanosleep,
[36] = __x64_sys_getitimer,
[37] = __x64_sys_alarm,
[38] = __x64_sys_setitimer,
[39] = __x64_sys_getpid,
[40] = __x64_sys_sendfile64,
[41] = __x64_sys_socket,
[42] = __x64_sys_connect,
[43] = __x64_sys_accept,
[44] = __x64_sys_sendto,
[45] = __x64_sys_recvfrom,
[46] = __x64_sys_sendmsg,
[47] = __x64_sys_recvmsg,
[48] = __x64_sys_shutdown,
[49] = __x64_sys_bind,
[50] = __x64_sys_listen,
[51] = __x64_sys_getsockname,
[52] = __x64_sys_getpeername,
[53] = __x64_sys_socketpair,
[54] = __x64_sys_setsockopt,
[55] = __x64_sys_getsockopt,
[56] = __x64_sys_clone,
[57] = __x64_sys_fork,
```

```
[58] = __x64_sys_vfork,
[59] = __x64_sys_execve,
[60] = __x64_sys_exit,
[61] = __x64_sys_wait4,
[62] = __x64_sys_kill,
[63] = __x64_sys_newuname,
[64] = __x64_sys_semget,
[65] = __x64_sys_semop,
[66] = __x64_sys_semctl,
[67] = __x64_sys_shmctl,
[68] = __x64_sys_msgget,
[69] = __x64_sys_msgsnd,
[70] = __x64_sys_msgrcv,
[71] = __x64_sys_msgctl,
[72] = __x64_sys_fcntl,
[73] = __x64_sys_flock,
[74] = __x64_sys_fsync,
[75] = __x64_sys_fdatasync,
[76] = __x64_sys_truncate,
[77] = __x64_sys_ftruncate,
[78] = __x64_sys_getdents,
[79] = __x64_sys_getcwd,
[80] = __x64_sys_chdir,
[81] = __x64_sys_fchdir,
[82] = __x64_sys_rename,
[83] = __x64_sys_mkdir,
[84] = __x64_sys_rmdir,
[85] = __x64_sys_creat,
[86] = __x64_sys_link,
[87] = __x64_sys_unlink,
[88] = __x64_sys_symlink,
[89] = __x64_sys_readlink,
[90] = __x64_sys_chmod,
[91] = __x64_sys_fchmod,
[92] = __x64_sys_chown,
[93] = __x64_sys_fchown,
[94] = __x64_sys_lchown,
[95] = __x64_sys_umask,
[96] = __x64_sys_gettimeofday,
[97] = __x64_sys_getrlimit,
[98] = __x64_sys_getrusage,
[99] = __x64_sys_sysinfo,
[100] = __x64_sys_times,
[101] = __x64_sys_ptrace,
[102] = __x64_sys_getuid,
[103] = __x64_sys_syslog,
[104] = __x64_sys_getgid,
[105] = __x64_sys_setuid,
[106] = __x64_sys_setgid,
[107] = __x64_sys_geteuid,
[108] = __x64_sys_getegid,
[109] = __x64_sys_setpgid,
[110] = __x64_sys_getppid,
[111] = __x64_sys_getpgrp,
[112] = __x64_sys_setsid,
[113] = __x64_sys_setreuid,
[114] = __x64_sys_setregid,
[115] = __x64_sys_getgroups,
[116] = __x64_sys_setgroups,
[117] = __x64_sys_setresuid,
[118] = __x64_sys_getresuid,
```

```
[119] = __x64_sys_setresgid,  
[120] = __x64_sys_getresgid,  
[121] = __x64_sys_getpgid,  
[122] = __x64_sys_setfsuid,  
[123] = __x64_sys_setfsgid,  
[124] = __x64_sys_getsid,  
[125] = __x64_sys_capget,  
[126] = __x64_sys_capset,  
[127] = __x64_sys_rt_sigpending,  
[128] = __x64_sys_rt_sigtimedwait,  
[129] = __x64_sys_rt_sigqueueinfo,  
[130] = __x64_sys_rt_sigsuspend,  
[131] = __x64_sys_sigaltstack,  
[132] = __x64_sys_utime,  
[133] = __x64_sys_mknod,  
[135] = __x64_sys_personality,  
[136] = __x64_sys_ustat,  
[137] = __x64_sys_statfs,  
[138] = __x64_sys_fstatfs,  
[139] = __x64_sys_sysfs,  
[140] = __x64_sys_getpriority,  
[141] = __x64_sys_setpriority,  
[142] = __x64_sys_sched_setparam,  
[143] = __x64_sys_sched_getparam,  
[144] = __x64_sys_sched_setscheduler,  
[145] = __x64_sys_sched_getscheduler,  
[146] = __x64_sys_sched_get_priority_max,  
[147] = __x64_sys_sched_get_priority_min,  
[148] = __x64_sys_sched_rr_get_interval,  
[149] = __x64_sys_mlock,  
[150] = __x64_sys_munlock,  
[151] = __x64_sys_mlockall,  
[152] = __x64_sys_munlockall,  
[153] = __x64_sys_vhangup,  
[154] = __x64_sys_modify_ldt,  
[155] = __x64_sys_pivot_root,  
[156] = __x64_sys_sysctl,  
[157] = __x64_sys_prctl,  
[158] = __x64_sys_arch_prctl,  
[159] = __x64_sys_adjtimex,  
[160] = __x64_sys_setrlimit,  
[161] = __x64_sys_chroot,  
[162] = __x64_sys_sync,  
[163] = __x64_sys_acct,  
[164] = __x64_sys_settimeofday,  
[165] = __x64_sys_mount,  
[166] = __x64_sys_umount,  
[167] = __x64_sys_swapon,  
[168] = __x64_sys_swapoff,  
[169] = __x64_sys_reboot,  
[170] = __x64_sys_sethostname,  
[171] = __x64_sys_setdomainname,  
[172] = __x64_sys_iopl,  
[173] = __x64_sys_ioperm,  
[175] = __x64_sys_init_module,  
[176] = __x64_sys_delete_module,  
[179] = __x64_sys_quotactl,  
[186] = __x64_sys_gettid,  
[187] = __x64_sys_readahead,  
[188] = __x64_sys_setxattr,  
[189] = __x64_sys_lsetxattr,
```

```
[190] = __x64_sys_fsetxattr,  
[191] = __x64_sys_getxattr,  
[192] = __x64_sys_lgetxattr,  
[193] = __x64_sys_fgetxattr,  
[194] = __x64_sys_listxattr,  
[195] = __x64_sys_llistxattr,  
[196] = __x64_sys_flistxattr,  
[197] = __x64_sys_removexattr,  
[198] = __x64_sys_lremovexattr,  
[199] = __x64_sys_fremovexattr,  
[200] = __x64_sys_tkill,  
[201] = __x64_sys_time,  
[202] = __x64_sys_futex,  
[203] = __x64_sys_sched_setaffinity,  
[204] = __x64_sys_sched_getaffinity,  
[206] = __x64_sys_io_setup,  
[207] = __x64_sys_io_destroy,  
[208] = __x64_sys_io_getevents,  
[209] = __x64_sys_io_submit,  
[210] = __x64_sys_io_cancel,  
[212] = __x64_sys_lookup_dcookie,  
[213] = __x64_sys_epoll_create,  
[216] = __x64_sys_remap_file_pages,  
[217] = __x64_sys_getdents64,  
[218] = __x64_sys_set_tid_address,  
[219] = __x64_sys_restart_syscall,  
[220] = __x64_sys_semtimedop,  
[221] = __x64_sys_fadvise64,  
[222] = __x64_sys_timer_create,  
[223] = __x64_sys_timer_settime,  
[224] = __x64_sys_timer_gettime,  
[225] = __x64_sys_timer_getoverrun,  
[226] = __x64_sys_timer_delete,  
[227] = __x64_sys_clock_settime,  
[228] = __x64_sys_clock_gettime,  
[229] = __x64_sys_clock_getres,  
[230] = __x64_sys_clock_nanosleep,  
[231] = __x64_sys_exit_group,  
[232] = __x64_sys_epoll_wait,  
[233] = __x64_sys_epoll_ctl,  
[234] = __x64_sys_tgkill,  
[235] = __x64_sys_utimes,  
[237] = __x64_sys_mbind,  
[238] = __x64_sys_set_mempolicy,  
[239] = __x64_sys_get_mempolicy,  
[240] = __x64_sys_mq_open,  
[241] = __x64_sys_mq_unlink,  
[242] = __x64_sys_mq_timedsend,  
[243] = __x64_sys_mq_timedreceive,  
[244] = __x64_sys_mq_notify,  
[245] = __x64_sys_mq_getsetattr,  
[246] = __x64_sys_kexec_load,  
[247] = __x64_sys_waitid,  
[248] = __x64_sys_add_key,  
[249] = __x64_sys_request_key,  
[250] = __x64_sys_keyctl,  
[251] = __x64_sys_ioprio_set,  
[252] = __x64_sys_ioprio_get,  
[253] = __x64_sys_inotify_init,  
[254] = __x64_sys_inotify_add_watch,  
[255] = __x64_sys_inotify_rm_watch,
```

```
[256] = __x64_sys_migrate_pages,  
[257] = __x64_sys_openat,  
[258] = __x64_sys_mkdirat,  
[259] = __x64_sys_mknodat,  
[260] = __x64_sys_fchownat,  
[261] = __x64_sys_futimesat,  
[262] = __x64_sys_newfstatat,  
[263] = __x64_sys_unlinkat,  
[264] = __x64_sys_renameat,  
[265] = __x64_sys_linkat,  
[266] = __x64_sys_symlinkat,  
[267] = __x64_sys_readlinkat,  
[268] = __x64_sys_fchmodat,  
[269] = __x64_sys_faccessat,  
[270] = __x64_sys_pselect6,  
[271] = __x64_sys_ppoll,  
[272] = __x64_sys_unshare,  
[273] = __x64_sys_set_robust_list,  
[274] = __x64_sys_get_robust_list,  
[275] = __x64_sys_splice,  
[276] = __x64_sys_tee,  
[277] = __x64_sys_sync_file_range,  
[278] = __x64_sys_vmsplice,  
[279] = __x64_sys_move_pages,  
[280] = __x64_sys_utimensat,  
[281] = __x64_sys_epoll_pwait,  
[282] = __x64_sys_signalfd,  
[283] = __x64_sys_timerfd_create,  
[284] = __x64_sys_eventfd,  
[285] = __x64_sys_fallocate,  
[286] = __x64_sys_timerfd_settime,  
[287] = __x64_sys_timerfd_gettime,  
[288] = __x64_sys_accept4,  
[289] = __x64_sys_signalfd4,  
[290] = __x64_sys_eventfd2,  
[291] = __x64_sys_epoll_create1,  
[292] = __x64_sys_dup3,  
[293] = __x64_sys_pipe2,  
[294] = __x64_sys_inotify_init1,  
[295] = __x64_sys_preadv,  
[296] = __x64_sys_pwritev,  
[297] = __x64_sys_rt_tgsigqueueinfo,  
[298] = __x64_sys_perf_event_open,  
[299] = __x64_sys_recvmmsg,  
[300] = __x64_sys_fanotify_init,  
[301] = __x64_sys_fanotify_mark,  
[302] = __x64_sys_prlimit64,  
[303] = __x64_sys_name_to_handle_at,  
[304] = __x64_sys_open_by_handle_at,  
[305] = __x64_sys_clock_adjtime,  
[306] = __x64_sys_syncfs,  
[307] = __x64_sys_sendmmsg,  
[308] = __x64_sys_setns,  
[309] = __x64_sys_getcpu,  
[310] = __x64_sys_process_vm_readv,  
[311] = __x64_sys_process_vm_writev,  
[312] = __x64_sys_kcmp,  
[313] = __x64_sys_finit_module,  
[314] = __x64_sys_sched_setattr,  
[315] = __x64_sys_sched_getattr,  
[316] = __x64_sys_renameat2,
```

```
[317] = __x64_sys_seccomp,  
[318] = __x64_sys_getrandom,  
[319] = __x64_sys_memfd_create,  
[320] = __x64_sys_kexec_file_load,  
[321] = __x64_sys_bpf,  
[322] = __x64_sys_execveat,  
[323] = __x64_sys_userfaultfd,  
[324] = __x64_sys_membarrier,  
[325] = __x64_sys_mlock2,  
[326] = __x64_sys_copy_file_range,  
[327] = __x64_sys_preadv2,  
[328] = __x64_sys_pwritev2,  
[329] = __x64_sys_pkey_mprotect,  
[330] = __x64_sys_pkey_alloc,  
[331] = __x64_sys_pkey_free,  
[332] = __x64_sys_statx,  
[333] = __x64_sys_io_pgetevents,  
[334] = __x64_sys_rseq,  
[424] = __x64_sys_pidfd_send_signal,  
[425] = __x64_sys_io_uring_setup,  
[426] = __x64_sys_io_uring_enter,  
[427] = __x64_sys_io_uring_register,  
[428] = __x64_sys_open_tree,  
[429] = __x64_sys_move_mount,  
[430] = __x64_sys_fsopen,  
[431] = __x64_sys_fsconfig,  
[432] = __x64_sys_fsmount,  
[433] = __x64_sys_fspick,  
[434] = __x64_sys_pidfd_open,  
[435] = __x64_sys_clone3,
```

PART 4

Task: Explain what is ELF file? Try readelf and objdump command on an ELF file, give screenshot of the output. Run an ELF file and cat /proc/PID/maps to give its memory layout.

Answer: ELF(Executable and Linkable Format) is used as the format of binary file, executable file, object code, shared library file, while it is main executable file format in Linux. Simply, ELF file is executable file in Linux.

As for the specific structure, ELF is a format for storing programs or fragments of programs on disk, created as a result of compiling and linking. An ELF file can be divided into sections.

For an executable program, these are the text section for the code, the data section for global variables and the rodata section that usually contains constant strings. The ELF file contains headers that give the description on how these sections should be stored in memory.

And I will give a diagram of ELF file below.

The diagram illustrates the mapping from an ELF File Image to an Executable in Memory. The ELF File Image is divided into sections: Elf Header, Text (Code) Header, Data Header, Text (Code), and Data. The Executable in Memory is divided into sections: Text (Code), Data, and a Stack (4096). Arrows indicate the mapping: Text (Code) from the ELF File Image to the Executable in Memory, and Data from the ELF File Image to the Executable in Memory.

Elf File Image	Executable in Memory
Elf Header	
Text (Code) Header	
Data Header	
Text (Code)	Text (Code)
Data	Data
	Stack (4096)

Labels for ELF File Image: text.offset, text.filesz, data.offset, data.filesz.

Labels for Executable in Memory: text.memsize, data.memsize, text.vaddr, data.vaddr.